Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Biology

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Gaseous Exchange

Learning Objectives

• Define breathing, respiration and gaseous exchange.
• Describe the human respiratory system in order.
• Explain the mechanism of breathing.
• Compare breathing in humans, fish and plants.

Key Concepts

• Breathing vs cellular respiration
• Air passage: nose, trachea, bronchi, alveoli
• Inhalation and exhalation: diaphragm and ribs
• Gaseous exchange in alveoli
• Disorders: asthma, bronchitis; smoking hazards

Teaching Methodology (Step by Step)

1. Ask students to feel their ribs move while breathing deeply.
2. Draw the respiratory system step by step with air flow arrows.
3. Demonstrate the bell-jar lung model.
4. Discuss smoking hazards with real health facts.

Classroom Activities

• Build a bell-jar lung model in groups.
• Measure breathing rate before and after exercise.

Assessment Questions

1. Name the air passage in order from nose to alveoli.
2. What is the role of the diaphragm in inhalation?
3. Where does gaseous exchange occur in humans?
4. Name two respiratory disorders.

Homework

• Draw and label the human respiratory system.
• Write the mechanism of breathing in steps.

Time Allocation

5 periods of 40 minutes each

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Unit 2: Homeostasis

Learning Objectives

• Define homeostasis and its importance.

• Explain osmoregulation in humans.

• Describe thermoregulation in the human body.

• Explain homeostasis in plants briefly.

Key Concepts

• Homeostasis: keeping internal conditions stable

• Osmoregulation: kidney structure and urine formation

• Thermoregulation: sweating, shivering, vasodilation

• Feedback mechanism concept

• Diabetes and kidney failure as imbalances

Teaching Methodology (Step by Step)

1. Ask why body temperature stays near 37 C in all seasons.
2. Draw the kidney and nephron simply; trace urine formation.
3. Explain sweating and shivering as cooling/heating systems.
4. Discuss diabetes: what fails in homeostasis?

Classroom Activities

• Draw and label the human urinary system.

• List body responses to hot and cold weather.

Assessment Questions

1. Define homeostasis.
2. Name the structural and functional unit of the kidney.
3. How does sweating cool the body?
4. What is osmoregulation?

Homework

• Draw the urinary system with labels.

• Write the steps of urine formation.

Time Allocation

5 periods of 40 minutes each

Unit 3: Coordination and Control

Learning Objectives

• Distinguish nervous and hormonal coordination.

• Describe the human nervous system divisions.

• Explain the reflex arc with a diagram.

• Describe major endocrine glands and their hormones.

Key Concepts

• Nervous vs hormonal control: speed and duration

• CNS: brain and spinal cord; PNS

• Neuron: structure and function

• Reflex arc: receptor to effector

• Glands: pituitary, thyroid, pancreas, adrenal

Teaching Methodology (Step by Step)

1. Touch a hot object story: the instant pull-back (reflex).
2. Draw the neuron large; trace an impulse path.
3. Act out the reflex arc with student volunteers.
4. Match glands with hormones in a quick game.

Classroom Activities

• Draw and label the reflex arc.

• Make a glands-hormones-functions chart.

Assessment Questions

1. Name the two divisions of the nervous system.
2. Draw a labelled neuron.
3. Which gland is called the master gland?
4. What is a reflex action? Give one example.

Homework

• Draw the reflex arc with labels.

• Write functions of insulin and thyroxine.

Time Allocation

6 periods of 40 minutes each

Unit 4: Support and Movement

Learning Objectives

• Describe the human skeletal system divisions.

• Explain the structure of bone and cartilage.

• Describe types of joints with examples.

• Explain muscle structure and movement.

Key Concepts

• Skeleton: axial and appendicular

• Bone vs cartilage

• Joints: fixed, hinge, ball-and-socket

• Muscles: antagonistic pairs (biceps/triceps)

• Disorders: arthritis, fracture first aid

Teaching Methodology (Step by Step)

1. Let students feel their elbow and shoulder joints move.
2. Show a skeleton chart; name major bones together.
3. Demonstrate antagonistic muscles by flexing the arm.
4. Discuss first aid for fractures.

Classroom Activities

• Label major bones on a skeleton diagram.

• Identify joint types in your own body.

Assessment Questions

1. Name the two divisions of the skeleton.
2. Give one example each of hinge and ball-and-socket joints.
3. Which muscles work as an antagonistic pair in the arm?
4. What is arthritis?

Homework

• Draw the human skeleton with labels.

• Write the functions of the skeletal system.

Time Allocation

5 periods of 40 minutes each

Unit 5: Reproduction

Learning Objectives

• Define asexual and sexual reproduction.

• Describe asexual methods in plants.

• Describe the human male and female reproductive systems.

• Explain the advantages of sexual reproduction.

Key Concepts

• Asexual: binary fission, budding, vegetative propagation

• Sexual reproduction: gametes and fertilization

• Flower structure: male and female parts

• Pollination and fertilization in plants

• Human reproductive systems overview

Teaching Methodology (Step by Step)

1. Show a flower; dissect and name its parts.
2. Compare asexual and sexual reproduction in a table.
3. Explain vegetative propagation with potato and stem examples.
4. Draw flower structure step by step.

Classroom Activities

• Dissect a flower and label its parts.

• Grow a plant from a stem cutting and observe.

Assessment Questions

1. Define vegetative propagation with one example.
2. Name the male and female parts of a flower.
3. What is pollination?
4. Give two advantages of sexual reproduction.

Homework

• Draw and label the flower structure.

• Write methods of asexual reproduction with examples.

Time Allocation

6 periods of 40 minutes each

Unit 6: Inheritance

Learning Objectives

• Define heredity, gene and allele.

• Explain Mendel's experiments simply.

• Solve monohybrid cross problems.

• Describe DNA structure and its role.

Key Concepts

• Heredity and variation

• Mendel: dominant and recessive traits

• Punnett square for monohybrid cross

• Genotype vs phenotype

• DNA: double helix; genes on chromosomes

Teaching Methodology (Step by Step)

1. Ask why children resemble parents: introduce heredity.
2. Tell Mendel's pea-plant story simply.
3. Solve Punnett squares together on the board.
4. Show the DNA double-helix model or diagram.

Classroom Activities

• Solve five monohybrid cross problems in pairs.

• Survey class traits (tongue rolling) and tally results.

Assessment Questions

1. Define gene and allele.
2. What is a dominant trait?
3. Complete a Punnett square: Tt x Tt.
4. Who is called the father of genetics?

Homework

• Solve the given genetics problems.

• Draw and label the DNA double helix.

Time Allocation

6 periods of 40 minutes each

Unit 7: Man and His Environment

Learning Objectives

• Define ecology and ecosystem.

• Describe biotic and abiotic components.

• Explain food chains and food webs.

• Discuss human impacts: pollution, deforestation.

Key Concepts

• Ecology, ecosystem, biosphere

• Producers, consumers, decomposers

• Food chain vs food web; trophic levels

• Pollution: air, water, land

• Conservation: 3Rs, afforestation

Teaching Methodology (Step by Step)

1. Build a food chain from local organisms together.
2. Draw a pond ecosystem with all components.
3. Discuss local pollution: causes students can see.
4. Plan a class tree-plantation or cleanliness drive.

Classroom Activities

• Draw a food web of a local habitat.

• List ten human actions harming the environment.

Assessment Questions

1. Define an ecosystem.
2. Give one food chain with four organisms.
3. Name three types of pollution.
4. What are the 3Rs of conservation?

Homework

• Draw a labelled food web.

• Write a paragraph on deforestation.

Time Allocation

5 periods of 40 minutes each

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Unit 8: Biotechnology

Learning Objectives

• Define biotechnology with examples.

• Explain fermentation and its products.

• Describe genetic engineering basics.

• Discuss benefits and concerns of biotechnology.

Key Concepts

• Biotechnology: using living things for products

• Fermentation: bread, yogurt, biogas

• Genetic engineering: inserting useful genes

• Insulin production by bacteria

• GM crops: benefits and debates

Teaching Methodology (Step by Step)

1. Start with bread and yogurt: everyday biotechnology.
2. Explain fermentation with a simple dough demo.
3. Tell the insulin story: from animals to engineered bacteria.
4. Debate GM crops: feed the world vs unknown risks.

Classroom Activities

• Observe dough rising; record time and temperature.

• List ten biotechnology products used at home.

Assessment Questions

1. Define biotechnology.
2. Name two products of fermentation.
3. How is insulin made by biotechnology?
4. Give one benefit and one concern of GM crops.

Homework

• Write the steps of insulin production.

• Write a paragraph on the uses of biotechnology.

Time Allocation

4 periods of 40 minutes each

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